Hariprasaath Sridhar

Python Engineer · FastAPI · React · AI/ML

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PROFESSIONAL SUMMARY

Python and AI/ML engineer with 1.5+ years of experience building full-stack monitoring platforms, REST backends and Generative AI applications. Delivered the T-SCADA alarm monitoring system for a metro rail project (SNMP, React, FastAPI, PostgreSQL), a training management portal and an HR analytics chatbot on a locally hosted LLM. Strong in core Python, applied statistics (Chi-Square, ANOVA) and classical ML, and currently building agentic workflows with LangGraph.

WORK EXPERIENCE

Engineer — Mar 2026 – Present

L&T Technology Services — Bengaluru, India · Promoted from Associate Engineer within 15 months

- **MIS Digitalisation – HR Dashboard:** built an AI-powered chatbot with Python, FastAPI and Ollama (Llama 3) to automate staff allocation and bench-related query resolution.
- Developed a training management portal (React JS, FastAPI, PostgreSQL) for BMRCL Metro O&M teams with Admin, Operator and Maintenance modules.
- Engineered REST APIs for content upload, quiz assessments and training data management, improving digital training access.
- Designed workforce analytics dashboards with React JS and FastAPI on SQLite and PostgreSQL for real-time data retrieval.

Associate Engineer — Dec 2024 – Mar 2026

L&T Technology Services — Bengaluru, India

- Designed and developed **T-SCADA**, a web-based alarm monitoring system using React, FastAPI and PostgreSQL, integrating CCTV, PIDS, PAS, Radio and networking subsystems for BMRCL Metro.
- Implemented SNMP-based alarm collection (traps and polling) for proactive fault detection and severity-based alerting.
- Built real-time monitoring dashboards and analytics components with React JS, FastAPI and PostgreSQL.
- Containerized and deployed backend services with Docker on Linux; supported testing and production deployments.

FEATURED PROJECT — T-SCADA ALARM MONITORING SOFTWARE

React JS · FastAPI · PostgreSQL · SNMP · Docker · Linux

- Web application that monitors field devices over the **SNMP protocol**, receiving traps and polling OIDs/MIBs from metro subsystems and converting them into structured alarm events.
- Designed the FastAPI backend and PostgreSQL schema for alarm storage, severity classification, acknowledgement and history, exposed through REST APIs.
- Built the React front end with live alarm lists, severity indicators and subsystem-wise status views for control-room operators.

TECHNICAL SKILLS

Languages	Python, SQL
Backend / Web	FastAPI, REST APIs, Pydantic, React JS, SNMP · PostgreSQL, SQLite
AI / GenAI	LangChain, LangGraph, RAG, Agentic AI, Prompt Engineering, FAISS, ChromaDB, Ollama, NLP
DevOps / Cloud	Docker, Git, GitHub, GitHub Actions, Linux, AWS Basics

CORE PYTHON

- **Data model & OOP:** classes, inheritance, dunder methods, dataclasses, abstract base classes.
- **Data structures:** list, tuple, dict, set; comprehensions, slicing, sorting with key functions.
- **Functional tools:** decorators, closures, lambda, map/filter/reduce, itertools, functools.
- **Iterators & generators:** lazy evaluation, `yield`, context managers (`with`).
- **Concurrency:** asyncio, threading, multiprocessing; understanding of the GIL.
- **Quality:** exception handling, type hints, logging, PEP 8, unit testing with pytest, virtual environments.

STATISTICS & MACHINE LEARNING

- **Chi-Square test:** feature selection for categorical variables and tests of independence; used to select features in the heart-disease project.
- **ANOVA:** F-test to compare group means and rank numeric features against a categorical target.
- **Descriptive & inferential statistics:** distributions, hypothesis testing, p-values, confidence intervals.
- **Linear algorithms:** Linear Regression and Logistic Regression with Scikit-learn, including regularization (Ridge/Lasso).
- **Model workflow:** feature engineering, PCA, train/test split, cross-validation, accuracy, precision/recall, F1, ROC-AUC.
- **Libraries:** NumPy, Pandas, Scikit-learn, TensorFlow, Matplotlib.

AI / ML PROJECTS

Enterprise Document Research Agent – LangGraph & RAG

Python, LangGraph, LangChain, RAG, FAISS, FastAPI, LLMs

- Built an agentic RAG system that routes queries, retrieves enterprise documents, evaluates context and generates grounded answers with source references.
- Implemented a LangGraph workflow (routing, retrieval, relevance grading, tool invocation, response generation) exposed through FastAPI REST APIs.

Text-to-SQL HR Analytics Agent – LangGraph & Ollama

Python, LangGraph, Ollama (Llama 3), SQLite, FastAPI, Streamlit

- Built an agent that converts natural-language HR questions (bench strength, staff allocation, skill search) into SQL and returns answers from the database, using Llama 3 hosted locally.
- Designed a flow with schema lookup, SQL generation, read-only validation, execution and automatic retry on query errors, served via FastAPI with a Streamlit interface.

Privacy-Preserving Heart Disease Prediction using ML & FHE

College Project

Python, Scikit-learn, Concrete ML, PCA, Google Colab

- Evaluated 9 ML algorithms across 4 feature-selection methods on the UCI Cleveland Heart Disease dataset; Logistic Regression with Chi-Square reached **93.44%** accuracy.
- Implemented Fully Homomorphic Encryption with Zama Concrete ML for inference on encrypted patient data (90.16% accuracy).

EDUCATION

B.Tech, Computer Science and Engineering

2020 – 2024

SASTRA University, Thanjavur, India · CGPA 7.4/10

Coursework: Computer Networks, Machine Learning, DBMS, Software Engineering

CERTIFICATIONS

Python for AI & ML · Claude 101 · AWS Solutions Architect (In Progress)